

- A ZKP for an R1CS is accomplished by converting the witness vector into finite field elliptic curve points.
- Also ZKP is non succinct using R1CS.

Encrypting our witness vector:

$$\text{Eg: } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4G_1 \\ 5G_2 \end{bmatrix} = \begin{bmatrix} 14G_1 \\ 32G_2 \end{bmatrix}$$

Notation: $[aG_1]_1$ is a G_1 elliptic curve pt.

$[aG_2]_2$ is a G_2 ec pt.

for L we encrypt w using G_1 .

So

$$La = \begin{bmatrix} l_{11} & \dots & l_{1m} \\ \vdots & & \vdots \\ l_{n1} & \dots & l_{nm} \end{bmatrix} \begin{bmatrix} [a_1 G_1]_1 \\ \vdots \\ [a_m G_1]_1 \end{bmatrix}$$
$$= \begin{bmatrix} \sum_{i=1}^m l_{1i} [a_i G_1]_1 \\ \vdots \\ \sum_{i=1}^m l_{ni} [a_i G_1]_1 \end{bmatrix}$$

Now the Hadamard product becomes
a list of elliptic curve pairings.

So we encrypt w using G_2 for use
in R .

$$Ra = \begin{bmatrix} r_{11} & \dots & r_{1m} \\ \vdots & & \vdots \\ r_{n1} & \dots & r_{nm} \end{bmatrix} \begin{bmatrix} [a_1 G_2]_2 \\ \vdots \\ [a_m G_2]_2 \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{i=1}^m r_{1i} [a_i h_2]_2 \\ \vdots \\ \sum_{i=1}^m r_{ni} [a_i h_2]_2 \end{bmatrix}$$

- Next step would be to encrypt a using G_{12} points so verifier can check for equality but as G_{12} pts are massive so we would rather have the verifier do the pairing of Oa elliptic curve points in G_1 with a G_2 point (multiplying by 1)

i.e.

$$(Oa)_{12} = \begin{bmatrix} \sum_1^m o_{1i} [a_i h_1]_1 \\ \vdots \\ \sum_1^m o_{ni} [a_i h_1]_1 \end{bmatrix} \circ \begin{bmatrix} G_2 \\ G_2 \\ \vdots \\ G_2 \end{bmatrix}$$

$\underbrace{\quad}_{\downarrow}$
pairing each pt.

- Public inputs? Just don't encrypt the public values in the witness.